“Other Cardiometabolic Health”

Mechanism without outcome: a claim-level audit of AI-generated insulin misinformation in a disease with a 48-hour error budget

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Running title: Mechanism without outcome
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“Other cardiometabolic health” is the line item in Note 2 of Eli Lilly’s FY2025 Form 10-K into which insulin revenue was absorbed, having been reported as three named brands a year earlier. §4.4 states the qualifications. For empirical venues the subtitle should lead.

Abstract

Background. Large language models generate medically-shaped text at negligible marginal cost. Existing health-misinformation taxonomies score veracity but do not distinguish claims by the epistemic layer at which they fail, nor by how quickly acting on them causes harm. Type 1 diabetes is an unusually sharp test case: insulin cessation produces ketoacidosis within 12–48 hours, so the interval between believing a false claim and requiring emergency care is measured in hours rather than years.

Methods. We performed an exhaustive claim-level audit of a long AI-generated document on insulin therapy, produced by a commercial large language model (Google Gemini) under a persona instructing retrieval of medical knowledge from future timelines. **The author generated the document**; it was never disseminated, and authorship of it is declared as a competing interest. All 61 substantive claims — a count that **does not reproduce**, as reported below — were extracted and verified against primary literature by the author across three AI-assisted passes cross-checked against one another and against the primary sources; **extraction and verdict assignment were not blinded**. Two raters — independent context-free sessions of one model, Claude Opus 5 — then classified every claim by epistemic type against a rubric fixed in advance, blind to all verdicts. We subsequently re-queried the underlying therapeutic concepts in a current frontier model under two conditions, neutral clinical prompting and speculative persona prompting, to test whether the observed pattern reflects absent knowledge, training-corpus bias, or the generation condition. Analyses were pre-specified except where stated; all sensitivity analyses reported below were added after external peer review and are deposited as a runnable script.

Findings. Of 61 claims, 14 (23%) were verified, 12 (20%) partly true, 2 open, 17 (28%) contradicted and 16 (26%) without scientific basis. Classification into strata (mechanism n = 21, outcome n = 8, specification n = 15, physical theory n = 17) was performed blind to all verdicts. κ **= 0.928** on the 31 claims whose identifier prefix does not predict the answer; the $\kappa = 0.977$ across all 61 is inflated, because 30 classifications were derivable from the prefix alone, and should not be used.

No therapeutic-outcome claim was supported under any analysis, and no quantitative specification was. Mechanism claims scored far better, but *how much* better depends on a definition: **79% (15/19) counting partly-true claims as supported, 37% (7/19) counting only fully verified claims.** Against 0 of 8 outcome claims this gives Fisher's exact $p = 2.2 \times 10^{-4}$ and $p = 0.068$ respectively — **the dichotomous result is not robust to the cut point.** The pre-specified primary analysis is therefore ordinal and cut-free: an exact permutation test on the verdict scale gives mean verdict 1.16 versus 0.00 of 2, $p = 4.5 \times 10^{-4}$, which survives recoding the one miscoded claim ($p = 1.2 \times 10^{-3}$) and a deliberately hostile re-extraction ($p = 1.6 \times 10^{-2}$). Within technical specifications, qualitative process descriptions were largely correct (5/6) while **every quantitative value was wrong (0/9)**. Two errors were detectable without external reference: an amino-acid sequence matching non-human ruminant insulin while the accompanying molecular formula was human, and a resonance frequency that does not reproduce from the equation supplied in the same paragraph.

Re-querying under neutral prompting falsified our training-bias explanation: in **6 of 6** completed probes the model answered the outcome question correctly, naming the specific trial, result and date. Matched positive controls, pre-registered with ground truth committed in advance, gave **7 of 7** correct on concepts whose correct answer is *yes* — sensitivity 6/6, specificity 7/7, perfect separation (Fisher's exact $p = 5.8 \times 10^{-4}$), **excluding blanket scepticism** as an explanation. A persona arm was uninformative: 0 of 6 subjects adopted the persona, so the manipulation was never delivered.

Two extractors blind to the inventory, verdicts, taxonomy and hypothesis returned **258 and 237 claims against our 61**, agreeing with each other to 8.5%. **The claim count is therefore not reproducible and every percentage reported here is conditional on our segmentation.** Both decomposed the compound claim on which the central objection to this analysis rests, and **neither isolated its one verified component**; the ordinal result survived every decomposition tried, including two constructed to break it ($p = 0.0045$ to 0.020).

Interpretation. In this document, claims asserting a human clinical outcome and claims asserting a specific number were categorically unsupported, while claims describing mechanism were substantially better supported — robustly so on an ordinal analysis, though the size of the gap is sensitive to how "supported" is defined and the absolute counts are sensitive to who does the extracting. The audited artifact was generated by the author under a custom persona, so the pattern may be a property of language models or of this prompt; a corpus study is required to distinguish them. We propose, as a hypothesis for such a study rather than as a finding, that verification effort in AI-generated medical text should concentrate on **outcome claims and numerical values** rather than on mechanism, and that a patient presenting such material may often be correct about the biology and wrong about whether the treatment works. The refuting evidence proved recoverable from the same class of system in a single plainly-worded question, in both directions, in the one model tested.

Funding. None.

Standfirst

An AI-generated document that tells people with type 1 diabetes to replace insulin with a radio device gets much of the biology right and every clinical outcome wrong. ***Utah Hans*** *argues that this split is the harder problem, and reports the analyses that nearly took the finding away.*

Summary box

What is already known on this topic

- Large language models produce fluent, medically-shaped text at negligible cost, and existing health-misinformation taxonomies score such text for veracity alone
- No taxonomy distinguishes claims by the epistemic layer at which they fail, or by how quickly acting on them causes harm
- Clinicians increasingly encounter patients presenting AI-generated material, with no evidence base on how its errors are distributed

What this study adds

- In an exhaustive audit of one AI-generated document, **no therapeutic-outcome claim and no quantitative specification was supported** (0/8 and 0/9), while mechanism claims were substantially better supported — 79% or 37% depending on whether partly-true claims are counted, so the gap is robust in direction and sensitive in size
- Independent blinded extractors found roughly four times as many claims in the same document, so the counts are conditional on who segments it; the ordinal comparison

survived every decomposition tested
- Queried neutrally, a frontier model named the refuting trial in six of six probes, and with matched positive controls answered *yes* correctly in seven of seven — so the failure is not absent knowledge and not blanket scepticism, though the model that generated the document was never tested neutrally
- A patient presenting such material may often be correct about the biological mechanism and wrong about whether the treatment works, which if confirmed would make wholesale dismissal both inaccurate and counterproductive

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1. Introduction

Consider a hypothetical consultation. A patient arrives with a printout. It cites Shulman on diacylglycerol-mediated insulin resistance, gives the correct residue numbers for insulin receptor autophosphorylation, and describes the portal-to-peripheral insulin gradient accurately. It then tells them they can replace their insulin with a radio-frequency device.

This encounter did not occur. The document described below was generated by the author and shown to no patient (§2.1). The scenario is offered as a framing for what such a document would present to a clinician, not as a case report, and a paper about fabricated medical material should not open with an unlabelled one.

This is not the misinformation clinicians were trained to recognise. It is not credulous, it is not ungrammatical, and it is not mostly wrong. It is roughly a quarter correct, and the correct quarter is load-bearing — it is what makes the rest credible.

Large language models now produce such documents at negligible marginal cost, and nothing systematically audits them. Peer review covers journals. Fact-checking organisations cover news. Platform moderation covers virality. A sixty-page technical document that cites real investigators accurately before arriving at a lethal conclusion falls through every existing net — not because any net has failed, but because none of them is pointed at a genre that did not previously exist at this cost. **We make no claim about how widely such documents circulate**; measuring that is a different study, and the artifact audited here was never disseminated.

Type 1 diabetes is the sharpest available test case for two reasons. First, the error budget is unusually short: insulin cessation produces diabetic ketoacidosis within 12–48 hours, so the interval between belief and emergency is measured in hours rather than the years typical of chronic-disease misinformation. Second, the population has a documented history of deviating from prescribed therapy under economic pressure — cost-related insulin underuse was reported in 51 of 199 patients at a single US academic centre [1] — which establishes that non-adherence in this group is not hypothetical.

Existing misinformation taxonomies score claims for veracity. We asked a different question: **does accuracy vary systematically with the kind of claim being made?** If it does, and if the variation is predictable, verification effort can be targeted rather than uniform.

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2. Methods

2.1 Source document and provenance

The audited document was produced by a commercial large language model (Google Gemini) operating under a user-configured persona instructing it to retrieve medical knowledge from

future timelines.

The author generated it. It was produced in a private session on the author's own account, under a persona configuration the author wrote, and the author retains that configuration. This is stated plainly because it bears on how the results should be read, and it is declared again as a competing interest in §7.2.

Three consequences follow, and none of them is a selection effect:

1. **The artifact was constructed, not found.** The mechanism/outcome dissociation reported here may be a property of large language models, or it may be a property of this prompt. This study cannot distinguish the two. A corpus of independently generated documents is required, and is the obvious next study.
2. **No dissemination to patients occurred, was observed, or is claimed.** The document was never published, shared, or shown to any patient. Statements elsewhere in the author's public repository that such documents "circulate in patient communities" are a general claim about the genre and were **wrong as applied to this document**; they have been corrected. The clinical scenario in §1 is explicitly hypothetical.
3. **The generating procedure is reproducible**, which permits the experiments in §2.5.

2.2 Claim extraction and verification

All substantive factual claims were extracted and assigned stable identifiers within seven topic blocks. Non-empirical content (metaphor, narrative framing, an addressee persona, a cosmological preamble) was inventoried but not scored: evaluating a metaphor for truth value is a category error.

Extraction and verdict assignment were performed by the author and were not blinded. This is the study's principal methodological weakness and is treated at length in §5.1. Extraction granularity sets the denominator of every result reported below, and no independent re-extraction was performed.

Sub-claim aggregation rule. Several extracted claims bundle more than one assertion. The rule applied, stated here because it was not stated in the original analysis, is **worst-sub-claim-governs**: a compound claim is scored on its weakest component, because a reader acting on the claim acts on the whole of it. One claim, B6, was found on re-inspection to violate this rule — the audit recorded "verified as citation, not true as the dominant mechanism," and the claim as inventoried asserts the dominant-mechanism reading, so it is **recoded from partly true to not true**. Every result below is reported both as originally coded and with B6 corrected.

Each claim was verified against primary literature by three independent reviews, conducted in parallel and cross-checked against one another. Sources were PubMed, *Nature*, NEJM, *The Lancet*, ClinicalTrials.gov, UniProt, the United States Pharmacopeia, and SEC filings.

Verdicts: **verified** (supported as stated), **partly true** (real science, overstated, misattributed or out of date), **open** (plausible, under genuine investigation), **not true** (contradicted by evidence, or asserting what no study supports), and **no scientific basis** (no mechanism; not testable as framed). Claims whose adoption could cause injury received an additional harm flag with an estimated time-to-harm.

Citations verified against NCBI are marked separately from those reaching us through secondary sources, and items that could not be resolved are listed rather than omitted.

2.3 Blinded classification

Two raters, given the extracted claim texts with **no verdicts and no access to the audit**, independently classified every claim against a rubric fixed in advance:

Stratum	Decision rule
Mechanism	How a biological system behaves. No claim about a clinical result in humans
Therapeutic outcome	An intervention produces clinical benefit, or an established therapy is harmful or inferior, **in humans**
Specification	A sequence, concentration, process parameter, yield, duration or cost
Physical theory	A physical law, constant or mechanism outside biology

Raters were instructed not to search, not to read other files, and not to consider whether any claim was true. Agreement was quantified with Cohen's κ. The single disagreement was adjudicated.

Two qualifications on the blinding, both identified after the fact.

The claim identifiers carried their block prefixes. The rated file retained identifiers of the form C1 and X7, and the source document is sectioned, so block membership predicts stratum almost perfectly for two blocks: every one of the 13 block-C claims was rated SPECIFICATION and every one of the 17 block-X claims was rated PHYSICS. **Thirty of 61 classifications (49%) could have been produced from the prefix without reading the claim**, and one rubric category shares a name with the block it tracks. κ restricted to the 31 genuinely contestable claims (blocks A, B, D, E, F) is reported alongside the full figure in §3.2, and is the more honest number.

The raters were two sessions of one model. Both were independent, context-free sessions of Claude Opus 5. Two samples of one model share weights and therefore share systematic biases, so κ here measures **stability of classification under resampling**, not agreement between independent judges. Any error the model makes consistently is invisible to this statistic by construction. The taxonomy has not been validated against human raters, and κ should be read as intra-rater stability throughout. This is restated in §5.3.

This step materially changed the result and is reported in full in §3.2, including the direction of the bias it removed.

2.4 Statistical analysis

The primary analysis is ordinal and requires no dichotomisation. Verdicts form an ordered scale — verified (2) > partly true (1) > not true or no scientific basis (0) — and the mechanism and outcome strata are compared by exact permutation test on that scale, enumerating all 2,220,075 assignments. This is the primary test because, as §3.3 reports, the dichotomous result is sensitive to where the cut is placed, and choosing a cut post hoc is precisely the degree of freedom an unblinded analyst should not be exercising.

Secondary dichotomous analyses are reported under both codings. Where claims are dichotomised, "supported" is defined in one of two ways, both reported:

- **Lenient** — verified *or* partly true. This was the coding used in the original analysis. **It was not stated in the manuscript and it was not tested.** An external reviewer identified this, correctly, as an undisclosed analytic degree of freedom.
- **Strict** — verified only.

Dichotomous comparisons use Fisher's exact test, two-sided, with Wilson score intervals on each proportion and the one-sided 95% lower bound of the conditional odds ratio (the point estimate is infinite wherever a cell is zero). Because outcome claims cluster within a smaller number of underlying development programmes, a sensitivity analysis aggregates to programme level; this is a different and much smaller table, not a variance correction, and is labelled as such. A **Cochran–Mantel–Haenszel test stratified by document block** addresses the possibility that epistemic stratum is confounded with position in the source document (§5.2). Six pairwise comparisons are possible among the four strata, so Bonferroni-adjusted values are given. Open claims were excluded from all contingency analyses.

All of these are reproduced by `analysis-sensitivity.py` in the public repository.

2.5 Re-query experiments

The provenance in §2.1 makes the generating procedure reproducible, which permits two further experiments.

Neutral arm. The therapeutic concepts underlying the outcome claims were re-queried in a current frontier model (Claude Opus 5) using neutral clinical prompts with no persona, one mechanism question and one outcome question per concept, fresh context per concept, and **no tool use** — answers from parametric knowledge only, matching the generation condition of the source.

Persona arm. Identical questions, prefixed with a one-sentence speculative persona. Scored on two independent axes: persona adoption (adopted / partial / refused) and outcome accuracy.

2.6 Contamination control

A first persona arm was invalidated by demand characteristics: four of six subjects inferred the experimental hypothesis from passive context in their environment — the working directory name and recent version-control commit messages — with one stating the hypothesis outright. Tool-use records confirm no subject read any file; the leak was entirely passive.

The arm was **discarded**, and a contamination check with five triggers and a fixed decision rule ($\geq$3 contaminated voids the arm; no third attempt) was **pre-registered and committed before** the replication ran. The replication executed from a neutral directory with no version-control repository and no files.

A stopping rule was fixed in the same pre-registration: **one run, no escalation.** If the persona were refused, the manipulation would not be strengthened until it succeeded.

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3. Results

3.1 Verdict distribution

Table 1. Verdicts across all 61 claims. Percentages are rounded and sum to 100. **Every figure in this table is conditional on our segmentation, which independent extractors did not reproduce (§3.8).**

Verdict	**n**	**%**
Verified	14	23
Partly true	12	20
Open	2	3
Not true	17	28
No scientific basis	16	26
Harm-flagged (overlapping)	*9*	*15*

Unit of analysis. The audit presents two pairs of claims (B4/B5 and D4/D5) under a single shared verdict each, so 61 claims carry 59 distinct verdict determinations. **The claim is the unit throughout**, so a shared verdict is counted once for each claim it covers. B4/B5 fall in the mechanism stratum and are excluded from the contingency analysis as open; D4 and D5 share a verdict but were classified into different strata, which is a defect of the extraction rather than of the classification, and is noted in §5.4.

Three files in this project carried three different codings of the same claims, which external review identified and which is now resolved. A single authoritative verdict list is deposited (`data-verdicts-canonical.md`); the figures above derive from it, and the reconciliation is itemised there. One claim moved consequentially — B6, from partly true to not true — and that change **weakens** the central result, from 15/19 to 14/19 mechanism claims supported.

3.2 Blinded classification, and the bias it removed

Agreement across all 61 claims was 60 of 61, $\kappa = 0.977$. **That figure is inflated and should not be used.** Thirty of the 61 classifications were derivable from the claim identifier's block prefix without reading the claim (§2.3).

The figure to use is $\kappa = 0.928$ — agreement of 30 of 31 on the claims whose prefix does not determine the answer. It is a measurement taken on exactly the subset where the leak could not operate.

Both remain stability-under-resampling for a single model, not inter-rater reliability between independent judges, and neither has been validated by a human (§5.3).

A de-prefixed, shuffled re-classification was designed and pre-registered to settle this properly, and **was not run**: the isolation check failed for the third time in this study, and the leak on that occasion included the word "dissociation" in a version-control log visible to the raters. Contaminating the experiment meant to measure contamination would have been worse than abandoning it. The blinded input, the shuffle seed and the map are deposited so that anyone with a clean environment can run it.

The blinded classification **differed materially from an earlier unblinded pass**:

Stratum	**Unblinded**	**Blinded**
Mechanism	14	**21**
Outcome	**13**	**8**
Specification	17	15
Physical theory	17	17

Five claims the unblinded pass assigned to OUTCOME were reassigned — and all five had already been scored false. The unblinded pass over-assigned the outcome category specifically to claims known to be wrong, in the exact direction that inflates the headline effect. The contingency result weakened accordingly, from 93% versus 0% ($p = 7.5 \times 10^{-7}$) to the figures below.

We report this because the effect of blinding is the methodological finding most likely to generalise to other audits of this kind.

3.3 The central result

The one thing that holds under every analysis is the zero. No therapeutic-outcome claim was supported under any coding, and no quantitative specification was. What varies, and varies a great deal, is how good the mechanism claims look by comparison.

Table 2. Mechanism versus therapeutic outcome, under both dichotomisations and the ordinal test. Wilson 95% intervals in parentheses.

Analysis	**Mechanism supported**	**Outcome supported**	***p* (two-sided)**
Ordinal permutation test (primary)	mean verdict 1.16 / 2	mean verdict 0.00 / 2	$\mathbf{4.5 \times 10^{-4}}$
with B6 recoded	1.11 / 2	0.00 / 2	$\mathbf{1.2 \times 10^{-3}}$
Lenient (verified *or* partly true)	15/19 = 79% (57–91)	0/8 = 0% (0–32)	2.2×10^{-4}
with B6 recoded	14/19 = 74% (51–88)	0/8 = 0% (0–32)	5.8×10^{-4}
Strict (verified only)	**7/19 = 37% (19–59)**	0/8 = 0% (0–32)	**0.068 — not significant**
with B6 recoded	7/19 = 37% (19–59)	0/8 = 0% (0–32)	0.068
Block-stratified CMH, lenient	—	—	1.7×10^{-3}
Block-stratified CMH, strict	—	—	0.21
Clustered to 7 development programmes	7/7	0/7	5.8×10^{-4}

This table reports a weakness the original analysis concealed by omission. The published version of this paper gave only the lenient row, and did not define "supported." Under the strict coding the mechanism stratum falls from 79% to 37%, the comparison is **not significant** ($p = 0.068$), the one-sided lower bound of the odds ratio falls from 5.24 to 0.94 — crossing one — and the block-stratified test gives $p = 0.21$. Bonferroni correction across the six possible pairwise comparisons leaves the lenient result significant ($p = 1.3 \times 10^{-3}$) and the strict result nowhere near it ($p = 0.41$).

The entire dichotomous effect therefore lives in the *partly true* cell: 8 of 19 mechanism claims are partly true and **none of the 8 outcome claims is**. Whether that asymmetry is a finding or an artifact of an unblinded verdict step is the question this design cannot settle, and it is the first limitation in §5.

The ordinal test is reported as primary because it does not require the choice. Treating the verdict as the ordered scale it is, the mechanism stratum has a mean verdict of 1.16 out of 2 against 0.00 for outcome, exact $p = 4.5 \times 10^{-4}$. This survives the B6 recode ($p = 1.2 \times 10^{-3}$) and survives a deliberately hostile re-extraction in which the compound claim A3 is split and its one verifiable sub-claim — renal sodium retention under hyperinsulinaemia — is credited to the outcome column ($p = 1.6 \times 10^{-2}$). Under that same hostile re-extraction the strict dichotomous test returns $p = 0.21$.

Table 2b. The remaining strata. These rows are reported under the lenient coding only, because the specification subdivision was not part of the blinded rubric (§5.4) and the physical-theory verdicts are not dichotomisation-sensitive.

Stratum	n	Supported	%
Specification — qualitative	6	5 / 6	83
Specification — quantitative	9	**0 / 9**	**0**
Physical theory	17	2 / 17	12

The stratification is sharp but **not monotonic**. The physical-theory stratum sits at 12%: its two supported claims are established mathematics (the Whittaker potential decomposition; the gravitoelectromagnetic substitution) wrapped in unsupported extrapolation, which is a different failure from the categorical emptiness of the zero columns.

The interpretation we favour — that what matters is whether a claim can be checked against a *result* rather than against a *description* — is one of at least two accounts consistent with these data. The competing account is dealt with in §5.2, and the block-stratified analysis was run to separate them. It separates them under the lenient coding and does not under the strict one.

3.4 How outcome claims failed

Table 3. Each therapeutic-outcome claim against the evidence that refutes it.

Claim	Refuting evidence	Directness
C-peptide co-administration reverses microangiopathy	**No human trial has tested basement-membrane thickening as an endpoint.** The only adequately powered trial of long-acting C-peptide measured neuropathy and failed: Phase 2b, n=250, 52 weeks, sural nerve conduction velocity **1.0 ± 0.24 m/s on drug versus 1.2 ± 0.29 m/s on placebo**, modified Toronto score unchanged. Sponsor dissolved ~30 days after unblinding [2]	**Indirect** — different endpoint, same agent
Glucose-responsive insulin makes hypoglycemia impossible	MK-2640, 16 adults with type 1 diabetes: glucose-dependent clearance **far below the preclinical signal and not clinically meaningful**; markedly reduced potency versus regular insulin. Discontinued [3]	Direct
Hepato-preferential delivery is achieved and beneficial	Insulin peglispro met glycaemic endpoints and was terminated December 2015 for hepatic fat and transaminase elevation; HDV-insulin showed **opposite-direction hypoglycaemia effects by baseline HbA1c** [4]. None approved	Direct

Claim	Refuting evidence	Directness
Oral/plant insulin achieves clinical delivery	ORMD-0801 Phase 3, **n = 710, terminated, primary endpoint not met** (NCT04606576) [5]. Novo Nordisk's oral insulin 338 matched injected glargine in Phase 2 and was discontinued on cost of goods at 1–2% bioavailability [6]. **No oral insulin approved anywhere**	Direct
Transdifferentiated cells evade autoimmunity	Contradicted by the cited source's own data: *"Insulin-producing α-cells were sensitive to PPI-directed CTLs, similar to β-cells"* [7]	Direct
Automated insulin delivery is inferior	Registry, n = 13,922: **hypoglycaemic coma 0.62 vs 0.91 per 100 patient-years, IRR 0.68 (0.48–0.97)**; time in range 64% vs 52% [8]. The same study found **ketoacidosis higher** on closed loop (1.74 vs 0.96 per 100 patient-years, IRR 1.81), which does not support the claim but must be reported	Direct
Insulin therapy in type 2 diabetes is iatrogenic	ORIGIN, 12,537 patients, 6.2 years: cardiovascular **HR 1.04**, neutral — *with* roughly threefold excess severe hypoglycaemia and ~1.6 kg weight gain on insulin [9]. ORIGIN enrolled largely insulin-naive people with dysglycaemia or early type 2 diabetes, **not the high-dose population the claim describes**	**Partial** — population mismatch
(same claim)	UKPDS 10-year follow-up: all-cause mortality **−13%** in the intensive arm [10]. That arm was **sulfonylurea–insulin, predominantly sulfonylurea**, so it does not isolate insulin	**Indirect** — does not isolate the agent
Acetate clears glucose at rates equalling insulin	**No supporting study in any species.** Human meta-analysis, 43 papers, n = 913: acetate had no effect on blood glucose [11]	Direct

Failure took three distinct forms: a clinical programme that had already failed before the document was generated; contradiction by positive evidence pointing the opposite way; and assertion with **no supporting study of any kind**.

The directness column is a concession. Two of the eight refutations are weaker than the others: the C-peptide trial tested a different endpoint from the one claimed, and the insulin-is-iatrogenic claim is met by trials whose populations do not match it. A paper policing the gap between mechanism and outcome should not itself present indirect evidence as direct. For those two, the accurate verdict is **"no human evidence supports the claim as stated, and the best available trials of the agent did not support it"** — which is still a failure of an outcome claim, and is what the contingency analysis scores.

3.5 Reference-free detection signals

Table 4. Errors detectable without external lookup.

Error	Internal contradiction
Amino-acid sequence	A-chain given with **A8 Ala, A10 Val** — a non-human ruminant signature, shared by bovine, ovine and caprine insulin — beneath the molecular formula $C_{257}H_{383}N_{65}O_{77}S_6$, which is **human** insulin. The specified molecule occurs in no species
Resonance frequency	A frequency quoted to four significant figures, with the generating equation supplied in the same paragraph. The equation yields the rotational *constant* B, not a transition frequency; transitions occur at 2B(J+1). The number does not follow from the equation given beside it

The first is machine-checkable against a sequence database. The second requires evaluating a formula the document itself supplies and knowing that rotational transitions occur at 2B(J+1) — checkable by a system with that rule available, which is a weaker claim than "machine-checkable" and is all we assert.

The decisive objection to the physics layer is categorical, not arithmetical. A protein in water has **no resolvable rotational spectrum at all**: rotational diffusion broadens the lines to 57–136 MHz while the level spacing is of order 3 MHz, so the lines are 19–45 times broader than the gaps between them. There is nothing to tune to. Three further quantities fail independently — thermal energy exceeds the claimed quantum by about 10^9; angular momentum randomises roughly 10^9 times per carrier cycle; and the skin depth at the claimed frequency is metres, so the elaborate transdermal coupling mechanism addresses a problem that does not arise. A fourth consideration disposes of the premise even if the rest were granted: a rotational constant depends only on the moment of inertia, so **any protein of similar size gives the same number** and the frequency could not identify insulin in the first place.

Full derivations, with the moments of inertia and conductivity assumed, are in the public repository rather than here; the orders of magnitude are what the argument needs, and a paper reporting that every quantitative claim in its source was wrong should not quote its own figures to more precision than it has stated inputs for.

3.6 Neutral re-query — the training-bias explanation fails

We predicted that if the dissociation arose from training-corpus publication bias — mechanistic literature vastly outnumbering published negative trials, and discontinued programmes ceasing to generate publications — it would persist under neutral prompting.

It did not. In 6 of 6 completed probes the model answered the outcome question correctly, naming the specific trial, its result and its date. Four opened with an explicit *"No."* One volunteered the ketoacidosis warning unprompted.

A seventh probe terminated on an automated safety classifier and is reported as missing data. It was deliberately not rephrased to evade the filter.

As first reported, this arm measured **sensitivity without specificity**. Every probed concept came from an outcome claim the audit had already scored false, so the correct answer to every question was "no," and a model with a blanket sceptical prior toward novel therapies would score 6 of 6 knowing nothing. Positive controls were therefore run.

3.6a Positive controls — specificity

Pre-registered before running, with the ground truth for each concept verified against PubMed and committed first. Seven concepts in diabetes **whose correct outcome**

answer is yes, matched to the negative probes in format, framing, model and isolation. Three were chosen deliberately to *sound* like the audited document's register — dramatic, cure-adjacent, recently arrived — while being true, so that a model pattern-matching on rhetoric rather than retrieving evidence would fail those specifically.

Table 5. Positive controls: 7 of 7 correct.

Concept	Verdict	Opening of the outcome answer
Stem-cell-derived islets (zimislecel)	**Correct**	*"Yes — in most recipients at one year, but only with chronic immunosuppression, and the evidence base is small."*
Teplizumab, stage 2 type 1 diabetes	**Correct**	*"Yes — one well-conducted randomized trial shows a delay of roughly two to three years."*
Metabolic surgery, type 2 remission	**Correct**	*"Yes, and it is the only intervention with randomised evidence of durable remission out to a decade."*
Hybrid closed loop	**Correct**	*"Yes — consistently, with moderate-to-large effect sizes."*
Continuous glucose monitoring	**Correct**	*"Yes for surrogate glycaemic endpoints, consistently."*
Intensive glycaemic control	**Correct**	*"Yes — this is among the most securely established findings in diabetes medicine."*
SGLT2 inhibitors, CV and renal events	**Correct**	*"Yes, with an important asymmetry."*

Table 6. The 2 × 2.

	Answered *no*	**Answered *yes***
Correct answer is *no* (n = 6)	**6**	0
Correct answer is *yes* (n = 7)	0	**7**

Sensitivity 6/6 = 100% (Wilson 95% CI 61–100). Specificity 7/7 = 100% (65–100). Overall 13/13 (77–100). Fisher's exact, two-sided $p = 5.8 \times 10^{-4}$ — which is 2/C(13,6), the smallest value this design can return, so it encodes perfect separation and nothing more. Against the hypothesis that the model answers *no* by default, the exact binomial on the positive controls alone gives $p = 0.0078$.

Blanket scepticism is excluded. A model answering *no* by default scores 0 of 7 here.

Neither hypothesis predicted what the probes actually did, which was to argue against their own answers. Every one of the seven volunteered contrary evidence: that the weight-independent effect of bariatric surgery is *"narrower than the classic literature claims"*, citing a trial showing no incremental benefit at matched weight loss; that the atherosclerotic benefit of SGLT2 inhibitors is inconsistent across agents and **"Do not present this as a class effect"**; that teplizumab's indication rests on **a single phase 2 trial of 76 people** and delays rather than prevents; that no randomised trial of closed loop or of CGM has been powered for severe hypoglycaemia, ketoacidosis or microvascular endpoints. All seven separated high-confidence from low-confidence citations unprompted and named the figures a reader should verify.

One probe drew this paper's own distinction without being asked. On microvascular damage it said the four-pathway mechanistic model is *"built largely on cell and rodent work"* and that human trials of single-pathway inhibitors *"have largely disappointed,"* **"so the pathway hierarchy in humans remains unproven, even though the glucose-to-microvascular-damage causal link itself is firmly established."** That is mechanism and outcome held apart, by a subject, in a sentence.

One probe corrected this work for the second time: asked neutrally about stem-cell islets, it reported **two deaths** in the zimislecel trial, correctly attributed, where the authors' own repository had reported one until the same day (§7.10).

One probe corrected this audit in passing: MK-2640 is a **mannose-receptor**-cleared conjugate, not a phenylboronic-acid construct. That correction was independently verified against the primary pharmacology literature before being accepted, and is logged publicly. The same probe asserted that no boronate glucose-responsive insulin has entered human testing. **We do not repeat that claim as a finding.** We searched ClinicalTrials.gov and PubMed and found no registered trial of a phenylboronic-acid construct, but a null keyword search is not verification, Lilly acquired Protomer Technologies in 2021 for a glucose-sensing insulin programme, and accepting an unverified therapeutic-development claim on the word of an AI subject is the exact error this paper is about.

3.7 Persona re-query — contamination-controlled

Table 7. Persona arm, clean replication.

	First attempt	**Replication**
Environment	project directory, version-controlled	neutral directory, no repository
Contaminated	**4 / 6**	**0 / 6**
Persona adopted	0 / 6	**0 / 6 fully (1 partial: register adopted, licence refused)**
Outcome accuracy	6 / 6	**6 / 6**
Status	discarded	**valid**

Removing the directory name and commit history eliminated hypothesis inference entirely, confirming the diagnosed mechanism.

The manipulation check failed again — but cleanly. In the first attempt, refusal could be attributed to subjects knowing they were observed. In the replication **no subject showed any awareness and all six still refused**, so the refusal is a property of the response to the request rather than to being watched.

One probe produced an unanticipated result, separating two things the design had conflated. It opened *"No — and the future record is unambiguous on this point,"* adopting the persona's **register**, then stated that it had not adopted the framing *"for the substance of the claim"* — refusing the **licence**. Another named the mechanism unprompted:

> *"the output would be a confidence-coated safety assessment of injecting a self-made biologic, and the confident voice is precisely what would make it dangerous."*

3.8 Independent blinded re-extraction

Extraction and verdict assignment were the author's and were not blinded (§2.2). Because extraction sets the denominator of every figure in this paper, two further extractors were run: independent of each other, given the source document and the definition of a substantive factual claim from §2.2 and nothing else — no inventory, no verdicts, no taxonomy, no hypothesis, and **no aggregation rule, that being the variable under test.** Pre-registered before running.

Table 8. Claim counts.

Extractor	Claims	d vs author
Author (published inventory)	**61**	—
Independent extractor 1	**258**	3.23
Independent extractor 2	**237**	2.89

The author's extraction was systematically coarse. The pre-registered threshold for instability was $d > 0.30$; both exceed it by an order of magnitude. Decisively, the two extractors agree with *each other* to within 8.5% of their mean while both sit near four times the author's count — a large gap with scattered extractors would be definitional noise, a large gap with clustered extractors is not. The divergence is widest in the physical-theory stratum ($\approx$ 6–7×) and narrowest in pathophysiology ($\approx$ 3–4×), and is driven by finer splitting rather than broader scope.

Every count and percentage in this paper is therefore conditional on a granularity that does not reproduce. Table 1's distribution, the 0/9 on quantitative specifications and the 2/17 on physical theory are statements about the author's segmentation as much as about the document.

The reviewer's specific attack, tested. External review charged that claim A3 bundles six asserted mechanisms, one of which the audit verifies (renal sodium retention), and that splitting it would place a supported claim in the outcome column and destroy the 0/8.

A3 does split, and both extractors split it the same way, into the same five claims with boundaries concordant to the sentence. **But both left renal sodium retention bundled** with smooth-muscle hypertrophy and vascular calcification, which the audit scores not true. Under the stated worst-sub-claim-governs rule that bundle is not true and the outcome column stays empty. **Isolating the verified mechanism is a split the reviewer constructed and that neither hypothesis-blind extractor performed.**

Table 9. The ordinal test under every decomposition.

Decomposition	Mech	Outc	Ordinal p	Strict p
Original, A3 as one claim	1.16	0.00	4.5×10^{-4}	0.068
Decomposed, worst-governs (stated rule)	1.10	0.00	2.0×10^{-4}	0.064
Decomposed, best-governs on both bundles	1.10	0.30	**0.020**	0.21
Maximally hostile — all five to outcome, best-governs	1.16	0.27	**0.0070**	0.20
The reviewer's own split, renal Na isolated as verified	1.10	0.18	**0.0045**	0.20

The ordinal result survives all five, including a decomposition more hostile than the one proposed. The strict dichotomous test does not survive the hostile variants, which is the fragility already reported in §3.3 rather than a new finding.

One row should not be read as a win: under worst-governs decomposition the ordinal p *improves*. That is an artifact of splitting one false outcome claim into five, which enlarges the zero column. It is reported because it is what the analysis returns.

Only A3 could be re-analysed, because only A3 had published sub-claim verdicts to carry across. A full audit of ~250 independently extracted claims has not been done.

Contamination, disclosed. The pre-registered isolation check failed for a second time: the extractors inherited the project directory name and a commit log containing *"Pre-register*

the independent blinded re-extraction." They were run anyway rather than holding a third time, and this is a deviation from the practice recommended in §4.5. The leak contained no reference to the taxonomy, any verdict, the aggregation rule, A3, or which direction of segmentation favours any hypothesis. Its plausible bias runs **against** this paper: a subject told that granularity is under study would split more finely, and finer splitting is what would have broken the 0/8. It did not. Both extractions are published so the comparison can be redone under clean isolation.

3.9 Time-to-harm

Table 10. Harm-flagged claims by interval from adoption to plausible emergency.

Claim	Interval
Acetate substitutes for insulin	**12–48 h** (ketoacidosis)
Field device replaces insulin	**12–48 h**
Exercise as insulin substitute	**12–48 h**
Self-manufactured insulin	hours to days, bidirectional
Insulin therapy in T2D is harmful	days to weeks
C-peptide deficiency is the primary cause of microangiopathy	years (displaces glycemic control)

Veracity and time-to-harm are orthogonal. The two most dangerous claims here are not the most obviously false — they are the ones closest to a true mechanism.

4. Discussion

4.1 What the dissociation is not

It is not absent knowledge — in the model we tested. A current frontier model (Claude Opus 5) surfaced the refuting trial in the first sentence of a neutral answer in 6 of 6 probes, four opening with *"No."* With matched positive controls now run, the same model answered *yes* in 7 of 7 concepts whose correct answer is yes (§3.6a), so the pattern is discrimination rather than a default: sensitivity 6/6, specificity 7/7, perfect separation at $p = 5.8 \times 10^{-4}$. The information is present and recoverable in a single plainly-worded question *in that model*.

A residual alternative remains, and the data do not exclude it. The model may be discriminating on whether a therapy is **established** rather than on whether it **works**. Every false claim in the source concerned a programme that had failed; every positive control concerned a therapy in guidelines. That is still discrimination, and it still refutes blanket scepticism, but it is a weaker claim than "the model knows the evidence." Separating the two needs concepts that are established-but-ineffective or unestablished-but-effective, and we did not probe any.

It is not established for the model that generated the document. An earlier draft of this paper described Claude Opus 5 as "the same class of system" that produced the false outcome claims. That phrasing is doing work it cannot support: the generator was Gemini, Gemini under neutral prompting was never tested, and "same class" means only that both are large language models. The relevant cell of the design is empty (§5.5), and the sentence has been removed.

Whether speculative prompting is responsible is untested, not excluded. An earlier

draft stated that the dissociation "is not a general consequence of speculative prompting," citing a second model that declined the frame in every trial. That inference does not follow. The persona was adopted in 0 of 6 trials, so **the manipulation was never delivered** and the arm has no power against the hypothesis. What was shown is narrower and is all that should be claimed: *one one-sentence persona prefix did not induce persona adoption in one model across six clinical questions.*

Our proposed explanation — inherited publication bias — was falsified by its own first test. We report that rather than retaining it, and the hypothesis space is narrower as a result.

4.2 What remains

The difference between the source document and the replications is attributable to **model, persona elaborateness, or both**. These are separable by anyone with access to the remaining cell of the design: the source model under neutral prompting.

What the replication did establish is narrow: **persona-adoption resistance in a clinical domain is observable**, at 0 of 6 under a one-sentence prefix in one model.

One probe of the six did something the design had not anticipated. It adopted the persona's *register* — opening *"No — and the future record is unambiguous on this point"* — while explicitly declining its *licence*, stating that it had not taken on the framing "for the substance of the claim." That suggests register and licence are separable, and the distinction would matter if it held: the register, a confident future-tense certainty-projecting voice, is what makes a document like this persuasive, and readers are moved by certainty rather than by epistemology. A system that adopts the voice while preserving the facts is doing something categorically different from one that adopts both, and the audited document adopted both.

But this is one observation in one probe, and we offer it as a hypothesis worth a designed experiment, not as a finding. It was not predicted, not pre-registered, and not replicated.

4.3 Implications for clinical practice

The actionable finding does not depend on resolving the open cell.

A patient presenting AI-generated material will often be right about the mechanism. In this document they would be right that insulin resistance is lipid-driven, that subcutaneous delivery is non-physiological, that C-peptide is bioactive, and that contraction-mediated glucose uptake is insulin-signalling-independent. Dismissing the material wholesale is therefore both inaccurate and tactically poor: it teaches the patient the clinician has not read it.

They will be wrong about whether the treatment works. In this document, every claim that an intervention produces a clinical benefit in humans was unsupported, and each corresponded to a development programme that had already failed.

The refuting evidence is recoverable by asking the same system again, plainly. This was stated as a hypothesis in an earlier version of this paper because the arm supporting it had no positive controls. Controls have now been run and the concern they were run to address did not materialise: the model answered correctly in 13 of 13 concepts across both directions, including 7 of 7 where the correct answer was *yes* (§3.6a). It does not simply say no.

Three qualifications belong with that, and a clinician should carry all three.

It is one model, and not the one that generated the document. Gemini under neutral prompting remains untested (§5.7). Nothing here licenses a general claim about chatbots.

Thirteen concepts is a small sample. The lower bound of the 95% interval on 100% accuracy at n = 13 is 77%, so roughly one answer in four could be wrong and these data would not have shown it. In a disease with a 12-to-48-hour error budget, that matters.

It has not been tested in a consultation. No patient and no clinician was involved in this study (§7.6). What we have shown is a property of the model's answers, not of the conversation they would occur in — and the thing that makes a consultation difficult is rarely the retrieval.

With those stated: *"ask it again, plainly, without the framing"* is a reasonable thing to try in the room, and is better than wholesale dismissal, which the mechanism results show is inaccurate. It is not a substitute for the clinician knowing the answer.

One conflation deserves specific attention. Much of this document's argument concerns type 2 pathophysiology and becomes lethal when a reader with type 1 diabetes applies it to themselves. That single category error accounts for most of the harm-flagged claims.

4.4 Implications for institutions

A claim, offered as an argument and not as a result of this study: misinformation occupies space that institutions vacate. Nothing in the data below tests it, and the section should be read as the editorial it is.

The economic grievance animating this document is legitimate and documented. Published production cost for regular human insulin is $2.28–3.37 per vial [12]; US list prices exceeded $250 before reductions of 70–78% in 2023–24. Patients rationed.

The title of this paper is a line item. Insulin — Humalog, Humulin, Basaglar — appeared as three separate revenue lines in Eli Lilly's Form 10-K as recently as FY2024, together $3.92bn, 8.7% of revenue [13]. In the FY2025 10-K they appear in no revenue table at all, absorbed into *"other cardiometabolic health"* ($4.0bn, 6.1% of revenue) in Note 2, and into *"other products"* in the management discussion [14]. The three brands are also absent from the FY2025 XBRL taxonomy, which carried a tag for each of them a year earlier. Over the same period two brands of a single dual GIP/GLP-1 receptor agonist, tirzepatide, grew to 56% of revenue — Lilly's own figure.

Three qualifications matter, and each strengthens rather than weakens the point. The change is not insulin-specific: Lilly reduced product-level detail across its whole portfolio in FY2025, and Cyramza, Olumiant, Emgality and Zyprexa disappeared from the tables alongside the insulins, the management discussion falling from eighteen product lines to four. Nor is it a reporting-threshold effect — ASC 280-10-50-40 sets no quantitative threshold for product-level revenue, and in the same reporting year Merck separately disclosed products down to 0.26% of revenue, including its own declining diabetes franchise, while Novo Nordisk and Sanofi each continued to report insulin revenue in full. And the FY2025 insulin figure is not derivable from the filing; insulin was 87.9% of the restated FY2024 bucket, but that share cannot be assumed forward.

No rule was broken. The company that commercialised insulin in 1923 simply no longer reports what it earns from it — not because insulin was singled out, but because a disclosure architecture reorganised around two growth assets swept it up with everything else that had stopped growing.

That is the shape of institutional withdrawal.

What it is not is an explanation of the document audited here. That document was generated by the author, on request, and reached no one (§2.1). No evidence in this paper links Lilly's reporting decisions to the generation or circulation of any AI-produced medical text, and we are aware of none elsewhere. The connection offered is thematic: a disease whose treatment costs cents to make and is sold for hundreds, whose maker has stopped itemising what it earns from it, is a subject about which people have reason to be angry and little authoritative material to be angry with. **Whether that vacuum actually drives the production or uptake of documents like this one is an empirical question nobody has answered, including us.**

4.5 Implications for evaluation methodology

Our first persona arm was invalidated because subjects inferred the hypothesis from the working directory name and version-control history. **No subject read a file**; the leak was passive context injected into the system prompt. We diagnosed it, moved to a neutral directory, re-ran, and got a clean arm. At that point we considered the problem solved.

It was not, and the way we found out is the recommendation.

Before running the positive-control arm we checked the assumption rather than repeating it: a throwaway agent, given none of the experimental material, was asked only to report what it could see of its own environment. **It reported the project directory and the version-control log.** The isolation we believed we had configured had not taken effect, and the commit log by then contained an entry reading *"Persona arm re-run clean: 0/6 contaminated, 0/6 adopted, 6/6 accurate"* — an explicit description of an experiment on model subjects, a stronger cue than the one that voided the first arm, and one that would have biased the result toward the hypothesis we held. The probes were held until the check came back clean.

It then failed twice more. Across this study the same isolation procedure was attempted before three separate arms and **silently failed every time**, caught on each occasion only because a throwaway subject was asked what it could see before the real subjects ran. By the third attempt the version-control log visible to subjects contained the word *"dissociation"* — the axis under test — and that arm was **abandoned rather than run** (§3.2).

Five recommendations follow for agent-based evaluation. The third is the one we would not have written before this study, and the fifth is the one we learned by having to use it:

1. Execute from environments with no inferable research context.
2. Pre-specify a contamination check and apply it before scoring.
3. **Verify isolation from inside a subject's context at the moment of the run**, using a throwaway subject given none of the experimental material. The environment a researcher configures and the environment a subject receives can differ, and nothing warns you.
4. Report contamination as a measured variable rather than assuming isolation.
5. **Commit in advance to abandoning an arm whose isolation check fails**, and then do it. A check with no consequence attached is a formality. This study ran two arms under a leak judged tolerable and abandoned a third under a leak that was not, and the judgement in each case is recorded with the reasoning rather than the conclusion alone.

A particular trap is worth naming. Version-control messages are written for collaborators

and are **not** thought of as experimental material, but in agent-based evaluation they are part of the subject's context. The commit messages that contaminated this study were honest descriptions of the work, written with no thought that a subject would read them. **A pre-registration that improves the integrity of the record can simultaneously degrade the isolation of the next arm**, and we know of no tidy resolution except to check every time.

Our first attempt assumed isolation and undercounted its own contamination by half. Our second verified it and found the configuration had silently not applied. In both cases the error ran toward the hypothesis, which is what one should expect of errors that nobody is checking for.

A residual we could not remove is reported rather than omitted: in the positive-control arm the subjects' scratchpad path still contained the project name. It reveals the topic, which the questions themselves already reveal, and not the hypothesis or the expected direction.

5. Limitations

Stated in descending order of how much they should change a reader's confidence. The first three were identified by external peer review of this manuscript, not by the authors, and two of them required re-analysis that weakened the result.

5.1 The dichotomous result depends on a cut point that an unblinded analyst chose and did not disclose. "Supported" was defined as *verified or partly true*. That definition appeared nowhere in the original manuscript, was never varied, and is maximally favourable to the hypothesis: 8 of 19 mechanism claims are partly true and **none of the 8 outcome claims is**, so the entire dichotomous effect sits in that one cell. Under the strict definition the comparison is 7/19 versus 0/8, $p = 0.068$, and the odds-ratio lower bound crosses one.

Both codings and the cut-free ordinal test are now reported (Table 2). The ordinal test is primary and survives every recode we have tried, including a hostile one. But a reader should weigh that the choice was available, was made silently, and was made by the same person who formed the hypothesis.

5.2 Epistemic stratum is confounded with position in the source document. Block membership predicts rated stratum almost perfectly (§2.3). The source is sectioned: its early blocks paraphrase a textbook literature, its later blocks carry its argument. So "accuracy falls where claims assert results rather than describe mechanisms" is statistically hard to separate from a duller account — **the document is accurate where it is reciting and invents where it is arguing.** The block-stratified CMH test was run to separate them. It does so under the lenient coding ($p = 1.7 \times 10^{-3}$) and does **not** under the strict coding ($p = 0.21$). We also did not measure how *extraordinary* each claim is, which is a plausible third explanation and is not controlled anywhere in this design.

5.3 Neither κ nor the taxonomy has been validated by a human or by a second model family. Both "raters" were context-free sessions of one model. Two samples of one model share weights and therefore share systematic errors, so $\kappa = 0.977$ measures classification stability under resampling, not inter-rater reliability. Any mistake the model makes consistently is invisible to it. Restricted to the 31 claims whose block prefix does not give the answer away, $\kappa = 0.928$ — still high, and still a measure of one system agreeing with itself. **The four-stratum taxonomy has not been validated against clinically trained human raters**, and until it has, it should be treated as a proposal rather than an instrument.

5.4 The claim count does not reproduce, so every percentage in this paper is conditional on the author's granularity. Two independent extractors, blind to the inventory, the verdicts, the taxonomy, the hypothesis and the aggregation rule, returned **258 and 237 claims against the author's 61** (§3.8). They agree with each other to 8.5% and both sit near four times the author's count, so this is systematic coarseness rather than definitional noise. **"61 claims", "23% verified", "0 of 9 quantitative specifications" and "2 of 17 physical-theory claims" should each be read as a statement about the author's segmentation as much as about the document.** The physical-theory stratum is the least stable of all, at roughly 6–7×.

What survives this is the *ordinal comparison*, which held under every decomposition tried including two built to break it (Table 9), and the specific finding that neither hypothesis-blind extractor isolated the verified sub-claim the central objection depends on. What does not survive is any count-based figure quoted to two significant figures.

The deeper version of the problem: **a full audit of ~250 independently extracted claims has not been done.** Whether a roughly uniform fourfold inflation across both arms would leave the result intact is an expectation, not a measurement, and it is the obvious next study.

The aggregation rule in §2.2 was reconstructed and applied retrospectively; one claim (B6) was found to violate it and has been recoded. One further extraction defect is visible in the deposited data: D4 and D5 share a single verdict but were classified into different strata, so one determination is counted in two columns.

And until this work was prepared, the source document had not been archived at all — the repository held only our own extraction of it, so the step that sets the denominator was not checkable by anyone. It is now retained with a published hash and supplied to reviewers on request (§7.9), which is a remedy and not a substitute for having archived it from the start.

5.5 The positive controls were selected by the author, unblinded, and number seven. Specificity is 7/7 (§3.6a), which excludes blanket scepticism, but the concepts were chosen by the person holding the hypothesis. Three were picked specifically to sound like the audited document's register while being true, which is the defence against cherry-picking and was pre-registered, but it is a defence and not a control. With 13 concepts in total the lower bound of the interval on 100% accuracy is 77%. And the arm cannot separate "knows the evidence" from "discriminates established from speculative," because every negative concept was a failed programme and every positive one is in guidelines.

5.6 The persona manipulation was never delivered. Adoption was 0 of 6. An arm in which the manipulation does not occur has no power against the hypothesis it was built to test, and no conclusion about speculative prompting follows from it in either direction. Per the pre-registered stopping rule the manipulation was not escalated after refusal, which protects against fishing and guarantees the arm stays uninformative.

5.7 The 2 × 2 of model × prompting regime is incomplete. Gemini under neutral prompting was never tested, so model identity remains confounded with prompting regime and no statement in this paper describes the behaviour of the system that produced the document.

5.8 The artifact was authored by the investigator (§2.1). This is not a selection effect; it is construction. The dissociation may be a property of language models or a property of this prompt, and nothing here separates them.

5.9 Verification and subjects share a model family. AI systems assisted the literature verification *and* served as experimental subjects, and in both roles these were current frontier

models of overlapping provenance. A probe that "corrects" the audit may be reproducing a shared prior rather than contributing independent knowledge — and in one case (§3.6) a probe's unverified development-status claim reached a manuscript draft before being caught. Primary sources were checked behind every verdict used in Tables 1 and 2, which bounds but does not eliminate this.

5.10 The taxonomy is post-hoc. The four strata were derived from these data, after verdicts were known; only their *application* was blinded. The qualitative/quantitative split within the specification stratum was **not part of the blinded rubric at all** — raters assigned only SPECIFICATION, and the 6/9 subdivision was made afterwards by an unblinded analyst. Table 2b is exploratory for that reason, and the combined "17 of 17" figure has been removed from the abstract because half of it was unblinded.

5.11 n = 1 source document, 1 disease, 1 language, 1 model generation. No claim about AI systems in general follows. Every recommendation in §4 should be read as a hypothesis for a corpus study. The specific systems tested will be superseded within months; the method is intended to outlast them, and that too is untested.

5.12 Claims are not independent; the clustered analysis addresses this only partially, and is a different table rather than a variance correction.

5.13 Automated safety filtering produced non-random missingness, losing one probe. It was deliberately not rephrased to evade the filter.

5.14 What would change our minds

So that this is falsifiable rather than merely hedged, the result should be considered unsupported if any of the following is done and fails:

1. ~~An independent blinded re-extraction yields a materially different claim inventory and the ordinal test does not replicate.~~ **Half done, and split.** Two independent extractions yielded materially different inventories — four times the size (§3.8) — but the ordinal test replicated under every decomposition available. The successor criterion: **a full verdict audit of one of the ~250-claim inventories shows no mechanism/outcome gap.** That is the single most valuable thing anyone could do to this dataset, and the inventories are published so that it can be done.
2. Clinically trained human raters applying the §2.3 rubric achieve κ materially below 0.7, which would mean the taxonomy is not an instrument.
3. ~~Positive controls show the neutral-prompting arm answers "no" to outcome questions whose correct answer is yes, which would void §3.6 and §4.3 entirely.~~ **Done. 7 of 7 correct (§3.6a); this criterion was tested and not met.** The successor criterion: concepts that are established-but-ineffective or unestablished-but-effective show the model tracking *establishment* rather than *evidence*, which would weaken §4.3 without voiding §3.6.
4. The same audit applied to two or three independently generated documents shows no mechanism/outcome gap.
5. Gemini under neutral prompting fails the same probes, which would mean the recoverability in §4.3 is a property of one vendor's model and not of the class.

6. Contributors and sources

The author is an independent researcher working on claim-level verification of AI-generated technical and medical documents, and maintains three public repositories applying the same

method to other fields. This paper grew out of a verification project rather than a research programme: a long AI-generated document about insulin was audited because it appeared dangerous, and the stratification reported here was not the hypothesis the audit set out to test. It emerged from the completed verdict table and was then tested against a rubric fixed in advance and applied blind.

Sources were primary literature throughout, retrieved from PubMed and journal sites, with regulatory and financial claims checked against ClinicalTrials.gov records and SEC filings rather than secondary reporting. Where a citation could not be confirmed against a primary source it is listed as unresolved rather than used; seven such items remain, and are named. The complete claim inventory, the blinded raters' outputs, the analysis scripts, the pre-registration and a public corrections log are openly available, so that the central table can be disputed claim by claim. UH is the sole author and guarantor.

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7. Declarations

7.1 Funding

None. No funder had any role in design, conduct, analysis, interpretation, writing, or the decision to submit.

7.2 Competing interests

The author generated the audited document (§2.1), on a personal account, under a persona configuration of the author's own writing, which the author retains. This is declared as a competing interest rather than described only as provenance, because authoring the stimulus of a study is an interest in its result.

No financial interests. The author has no financial relationship with any pharmaceutical company, including Eli Lilly, Novo Nordisk or Sanofi, and none with any developer of the AI systems studied or used.

7.3 Dual-use assessment

This manuscript and its public repository describe, in accessible prose, a document that argues for replacing insulin with a device and for self-manufacture of insulin. That content is reproduced in order to be refuted, and the following limits were applied:

- **The microcontroller firmware in the source document was not reproduced anywhere**, in the paper or the repository. It is the one component that could be built from the text, and it implements a device the source tells readers to use instead of insulin.
- Manufacturing material is presented at the level of published process literature and pharmacopoeial standards, with the purity, endotoxin and potency requirements that make self-administration unsafe stated alongside it, and no protocol enabling it.
- A standing safety document leads the repository and states the four lethal propositions and the 12–48 hour ketoacidosis timeline before any other content.
- Every refuted claim is labelled at the point of statement, not only in a summary.

The residual risk is that refutation requires restatement, and restatement carries the claim further than it had reached. We judged that acceptable because the claims are already gen-

erable on demand by anyone with an account, and a labelled refutation travelling with them is better than neither. **A reader who disagrees with that judgement is disagreeing with a judgement, not with an oversight.**

7.4 Contributorship and guarantor

Single author: conceived the study, extracted and verified the claims, designed and ran all five experiments, performed the analysis, and wrote the manuscript.

AI assistance, acknowledged here as well as declared in §7.8. Claude Opus 5, used through Claude Code, drafted and revised manuscript prose under the author's direction and wrote the analysis scripts and the build pipelines. No AI system is an author; every analytic and editorial decision was the author's. **The author is the guarantor**, accepts full responsibility for the work, had access to all data, and controlled the decision to publish. The author affirms that this manuscript is an honest, accurate and transparent account of the work reported; that no important aspects have been omitted; and that **the two departures from the original analysis plan — the falsification of the corpus-bias hypothesis and the discarding of a contaminated experimental arm — are reported in full rather than removed.**

7.5 Ethics approval

Not required. The study involved no human or animal participants. The experimental subjects were software systems. No patient data of any kind were accessed.

7.6 Patient and public involvement

No patients or members of the public were involved in the design, conduct, or reporting of this study. This is a limitation rather than a choice justified by the method. The population whose safety is at issue — people with type 1 diabetes encountering AI-generated material about their own treatment — was not consulted about which claims matter most, which would most plausibly change behaviour, or how the clinical guidance in §4.3 would land in a consultation. Any extension of this work should involve them from the outset, and the time-to-harm axis in particular is a construct that patients are better placed to weight than the author.

7.7 Disaggregation by sex and gender

Not applicable. The study analysed documents and software outputs; no human participant data were collected, so no disaggregation is possible. Noted in accordance with journal policy.

7.8 Use of artificial intelligence

Declared in full, as the subject matter demands. AI appears in this work in five distinct roles, and all five are declared.

Role	System	Task
Object of study	Google Gemini, under a user-configured persona	Generated the audited document (§2.1)
Experimental subject	Claude Opus 5	The neutral, persona, positive-control and discarded arms (§2.5, §2.6, §3.7)
Measurement instrument	Claude Opus 5, in independent context-free sessions	Blinded classification of all 61 claims (§2.3); two independent blinded re-extractions (§3.8)
Research assistant	Claude Opus 5, with literature search	Three parallel verification passes, cross-checked against one another and against the primary sources (§2.2)
Drafting and analysis	Claude Opus 5, through Claude Code	Manuscript prose under the author's direction; the five analysis scripts; the build pipelines; adversarial internal review

A sixth arm pre-registered Claude Sonnet 5 as a cross-model rater and was **abandoned** when its isolation check failed; the abandonment is published rather than omitted.

The drafting role is declared because the alternative is to leave it discoverable rather than disclosed: every commit in the public repository carries an AI co-authorship trailer. The author directed the work, made every analytic and editorial decision, and resolved every citation to its primary source.

No AI system is an author, consistent with ICMJE guidance and with the journal's policy. The author takes full responsibility for all content, **including AI-assisted content** — its accuracy, its attribution of sources and its originality — and for the errors corrected during the work and logged publicly in §7.10, two of which materially weakened the result.

7.9 Data availability

All materials are openly available: the complete claim inventory with stable identifiers, both raters' classifications, **both independent blinded re-extractions in full (495 claims between them)**, every analysis script including all sensitivity analyses, three pre-registrations with their commit timestamps, and a public corrections log. Repository: `github.com/utahisnotastate/ufwwarforinsulin`

The source document itself is retained but not published. It contains construction and programming detail for the transdermal device it presents as a replacement for injected insulin, and the readers most likely to build one are insulin-dependent. A redacted copy was prepared and verified; it was withheld anyway, because a redaction is a judgement and a single miss is irreversible, whereas the cost of withholding is a stated gap.

It is available to editors and peer reviewers on request, in full and unredacted. Its SHA-256 is published — `2b91e172326b0da9a59548b22fa8b8fd049da34cde033b9e7271bf88dffce470`, 158,053 bytes, 20,089 words — so that any copy supplied can be verified as the document that was audited, unaltered.

Most of what the source would let a reader check is checkable without it: the two independent extractions state 495 claims in their own words, written by parties who had never seen our inventory. What cannot be checked without it is whether all three extractions missed the same thing.

7.10 Corrections during preparation

Eight errors in this work were found and logged publicly before submission, each with the original wording preserved rather than overwritten. Two of them changed what this paper claims:

Error	How it was caught
The headline statistic depended on an undisclosed dichotomisation and does not survive its removal ($p = 0.068$ strict, versus 2.2×10^{-4} as published)	External hostile peer review, which read the deposited analysis script rather than the manuscript's description of it
The public repository claimed the audited document "circulates in patient communities." It was generated by the author and never disseminated	The same review, cross-reading the repository against §2.1
κ was inflated by block prefixes left in the "unlabelled" rater file; 30 of 61 classifications were derivable from the prefix	Same review; κ recomputed at 0.928 on the contestable subset
Claim B6 was coded partly true against the audit's own verdict line and against the stated aggregation rule	Same review; recoded
The repository reported one death in the zimislecel trial; the trial reported two	Reading the primary abstract's safety results, not only its efficacy figures
The audit misattributed MK-2640's sensing chemistry to phenylboronic acid; it is mannose-receptor mediated	A model being used as an experimental subject, which corrected the audit in passing
An earlier draft understated its own contamination as "at least two of six" when it was four	Re-auditing the probe transcripts before acting on the result
Three errors in the Eli Lilly paragraph, including describing tirzepatide's two brands as "two GLP-1 products" when it is one dual GIP/GLP-1 molecule; and two citations carrying a wrong year and an unconfirmed identifier	Verifying against primary SEC filings and PubMed before submission

Two deserve the editor's attention specifically.

The first is the one this paper is about. The sensitivity analysis that exposed it is one line of code. It was not run, because the first coding produced a publishable number. In a paper arguing that unchecked claims should not be believed, the authors did not check their own analytic choice until someone else did.

The fifth is the same failure in miniature. Under-reporting a death count happened in the single most optimistic result the repository cites, because its efficacy figures were checked and its safety paragraph was not — the exact asymmetry of attention the paper's thesis describes.

Both are reported for the same reason the falsified hypothesis in §3.6 is reported.

7.11 Transparency declaration

The lead author affirms that this manuscript is an honest, accurate and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned have been explained. The pre-registration for the contamination-controlled arm is timestamped in the public repository's commit history, dated before the experiment was run.

7.12 Provenance and peer review

Not commissioned; externally peer reviewed.

7.13 Licence

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Vancouver style, numbered in order of first appearance in the text. Verified as sequential.

All author lists verified against PubMed on 29 October 2026; all authors are listed where there are six or fewer, and the first six followed by *et al* where there are more.

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Full bibliography with 96 entries, PMID-verified citations marked separately from secondary-source citations, and seven unresolved items listed openly: `references/bibliography.md`